VOICEORCH / SNAPSERVE

Pre-Launch Regression Testing Checklist

A module-by-module manual-QA sign-off sheet for the AI voice-agent orchestration platform. Execute the full regression pass before go-live.

Document	Scope	Modules
Regression v1.0	**Manual QA**	**17 areas**

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How to use this checklist

Work top-to-bottom through each module. Tick the box (☐ → ☑) once a scenario passes on a clean build. Treat every P0 scenario as a hard launch gate — the platform does not ship until all P0 items are green. Record the tester, date, result, and any defect reference in the results grid at the end of each module so the run is auditable. Re-run the affected module after any fix and re-sign.

Priority legend

P0 — **Launch blocker**
Core revenue / call / auth / billing paths. Must be 100% green before launch.

P1 — **High**
Important features that degrade the product if broken but are not strictly call-blocking.

P2 — **Medium**
Secondary, admin-internal, or nice-to-have flows; verify but lower urgency.

Focus areas — executive summary

The seven highest-leverage risk areas. Concentrate the deepest testing effort here.

1. **Live voice call pipeline**
   STT→LLM→TTS, turn-taking, barge-in, latency — the heart of the product, highest risk.
2. **Authentication & tenant isolation**
   Google OAuth, sessions, and “user A cannot see user B’s data”.
3. **Billing & wallet accuracy**
   Managed vs BYOP rate resolution, per-minute charge, welcome credit, balance gating.
4. **Telephony & provider integrations**
   Inbound/outbound, webhooks, provider key handling, fallback chains.
5. **Scheduling end-to-end**
   Agent-booked + public-booked, Google Calendar sync, timezone correctness, double-booking prevention.
6. **Compliance**
   DPDP disclosure, PII redaction, data erasure.
7. **Admin operations**
   Managed keys, impersonation, usage monitoring, kill switches.

MODULE 1

Authentication & Onboarding

P0

P0 — Launch blocker

FEATURES

Google OAuth login, email/password login, registration, password reset (OTP), session JWT, first-time provisioning, logout.

TEST SCENARIOS

- [ ] Google OAuth sign-in succeeds and lands on dashboard with a valid session.
- [ ] Email/password registration creates an account; duplicate email is rejected.
- [ ] Email/password login succeeds with correct credentials; wrong password is rejected.
- [ ] Password-reset OTP is emailed, accepted once, expires, and is rate-limited against brute force.
- [ ] First-time user is provisioned with default keys + $5 welcome credit exactly once.
- [ ] Session persists across reload (localStorage token) and expires after TTL.
- [ ] SSE auth via ?token= query works for live streams.
- [ ] Logout clears the session and protected routes redirect to login.
- [ ] Expired/tampered JWT is rejected on protected API routes.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 2

Tenant Isolation & Authorization

P0

P0 — Launch blocker

FEATURES

Server-side ownership enforcement across all tenant resources.

TEST SCENARIOS

- [ ] User A cannot read/edit/delete User B’s agents, calls, providers, bookings, phone numbers, campaigns (try by ID).
- [ ] Call-control endpoints reject requests for calls the user does not own.
- [ ] Recording/transcript download is scoped to the owning tenant only.
- [ ] Admin routes reject requests without a valid x-admin-token.
- [ ] Non-admin user cannot reach any /api/admin* endpoint.

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Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 3

Dashboard / Overview

P1

P1 — High

FEATURES

Stats cards (agents, calls today, success rate, avg duration), call-volume chart, status pie, recent calls.

TEST SCENARIOS

- [ ] Stats reflect accurate counts for the logged-in tenant only.
- [ ] Empty-state renders cleanly for a brand-new account (no agents/calls).
- [ ] Call-volume chart shows the correct 14-day window.
- [ ] Status breakdown percentages sum correctly.
- [ ] Navigation to agents/calls from the dashboard works.

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Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 4

Agent Management

P0

P0 — Launch blocker

FEATURES

Create (builder wizard), edit, toggle active/inactive, delete, provider pipeline config (ASR/LLM/TTS/telephony), system prompt/greeting, variables, tools, knowledge-base attach, versions, templates.

TEST SCENARIOS

- ☐ Create agent via wizard with valid config saves successfully (no silent 400).
- ☐ Required-field validation blocks incomplete submissions with a clear error.
- ☐ Edit agent persists all fields (prompt, greeting, voice, providers, advanced settings).
- ☐ Toggle active/inactive updates status and reflects in list immediately.
- ☐ Delete agent removes it and associated UI references cleanly.
- ☐ Per-minute cost displays correctly per billing mode (managed vs BYOP).
- ☐ Variable injection ({{variables}}) resolves in prompt/greeting.
- ☐ Tool/function definitions save and are available at call time.
- ☐ Agent versioning records and can be viewed.
- ☐ Creating from a template prefills fields correctly.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 5

Live Voice Call Pipeline

P0

P0 — Launch blocker · highest risk

FEATURES

STT/ASR (Deepgram/Azure/Sarvam/Whisper/AssemblyAI), LLM streaming + fallback chain, TTS (ElevenLabs/Cartesia/OpenAI/Deepgram/Polly/Azure), endpointing/turn-taking, barge-in, gender detection/salutation, caller memory, warm transfer, AMD/voicemail, language handling, recording.

TEST SCENARIOS

- ☐ End-to-end web call: greeting plays, caller is heard, agent responds with low latency.
- ☐ STT streaming produces accurate transcripts; final transcript is captured even if only partials stream (force-finalize safety net).
- ☐ Stall backstop fires a recovery reprompt when no transcript arrives; call ends after max reprompts.
- ☐ LLM first-token latency is acceptable; streamed clauses flush to TTS smoothly.
- ☐ LLM fallback chain engages when the primary provider errors (no dropped call).
- ☐ TTS audio is clear, correctly encoded (mulaw for telephony), with no tail clipping of final words.
- ☐ Turn-taking: agent waits appropriately; digit-entry windows (OTP/policy ID) don’t cut the caller off early.
- ☐ Barge-in: caller can interrupt mid-sentence; agent’s own echo does not self-interrupt.
- ☐ Gender detection injects correct Sir/Ma’am salutation after greeting; “call me sir” manual override works.
- ☐ Caller memory: returning caller is recognized by canonical phone identity; dropped-call resume skips intro when appropriate (and cold-starts on garbage tails).
- ☐ Warm transfer to a human: whisper briefing is spoken, then bridged; caller not dropped (per-provider XML differences).
- ☐ AMD/voicemail: machine detected, message left after the beep.
- ☐ Language: Indic script auto-detection works; explicit language request latches; single-language rule doesn’t translate brand/proper nouns.
- ☐ Tool-call output never leaks raw tags to TTS (no spoken JSON/tags).
- ☐ Recording captures stereo (caller/agent) and is downloadable post-call.
- ☐ Pre-greeting drop resilience: setup failures fall back, don’t crash the call.

SIGN-OFF

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MODULE 6

Calls Log & Detail

P1

P1 — High

FEATURES

Filterable call history, search, CSV export, transcript view, audio playback, latency/token metadata, disposition.

TEST SCENARIOS

- [ ] Call list filters by status, agent, date range correctly.
- [ ] Search by phone number / call ID returns the right call.
- [ ] CSV export contains the filtered rows with correct columns.
- [ ] Call detail shows full transcript, recording playback, and metadata.
- [ ] Call-end flow auto-computes duration and updates agent success metrics.
- [ ] Disposition / outcome tagging displays correctly.

SIGN-OFF

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MODULE 7

Providers (BYOP)

P0

P0 — Launch blocker

FEATURES

Add/remove API keys grouped by ASR/LLM/TTS/Telephony, masked display, encryption at rest, regional settings.

TEST SCENARIOS

- ☐ Adding a provider key saves, encrypts at rest, and shows only the last-4 hint.
- ☐ Removing a key works and reverts the agent to managed billing if no keys remain.
- ☐ Invalid/empty key is rejected with a clear error.
- ☐ Adding any matching key for an agent’s pipeline switches that agent to BYOP rate.
- ☐ Region-specific providers (e.g. Azure) persist their region setting.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 8

Billing, Wallet & Usage

P0

P0 — Launch blocker

FEATURES

Managed ($0.05/min) vs BYOP ($0.035/min) dynamic rate resolution, wallet balance, top-ups, transaction history, usage tracking (STT/TTS/LLM tokens & chars), remaining-budget display.

TEST SCENARIOS

- ☐ Billing mode resolves correctly at call-end (BYOP if any matching key, else managed).
- ☐ Per-minute charge deducts the correct amount from the wallet.
- ☐ Welcome credit applied once; balance shows correctly.
- ☐ Low/zero balance blocks new calls with a clear message.
- ☐ Transaction history lists top-ups and call charges accurately.
- ☐ Usage page shows correct per-call tokens/characters per service.
- ☐ Refund of a specific call (admin) restores the correct amount.

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Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 9

Scheduling

P0

P0 — Launch blocker · for booking integrity

FEATURES

Calendars, event types, availability schedules, date overrides, agent-booked appointments (mid-call tools), public booking page (/book/:slug), Google Calendar sync (create/reschedule/cancel), Meet links, email notifications, webhooks, advisory-lock double-booking prevention.

TEST SCENARIOS

- ☐ Create event type, availability schedule, and override; slots reflect them.
- ☐ Public booking page lists correct open slots and books successfully.
- ☐ Agent books a slot mid-call; booking appears on the Scheduling page.
- ☐ Reschedule and cancel work from both public link and dashboard.
- ☐ Booking lands at the correct local time (timezone correctness — no UTC/offset bug).
- ☐ Double-booking the same slot concurrently is prevented (advisory lock).
- ☐ Google Calendar sync creates/updates/deletes the event; Meet link appears in email + public page.
- ☐ Confirmation/host-notification emails send (and gracefully skip when email env absent).
- ☐ Scheduling webhooks dispatch on booking events.

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Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 10

Phone Numbers & Telephony

P0

P0 — Launch blocker

FEATURES

Provisioning/managing DIDs, managed-number pool, inbound + outbound calls, provider webhooks (Twilio/Plivo/VoBiz), KYC-gated outbound, OTP verification for outbound.

TEST SCENARIOS

- ☐ Provision/manage a phone number; managed-number fallback works for users without their own.
- ☐ Outbound call initiation requires KYC/OTP where applicable and then connects.
- ☐ Inbound call routes to the correct agent.
- ☐ Provider webhooks are signature-verified and processed.
- ☐ Outbound caller ID uses a valid provisioned number (not the customer’s number).

SIGN-OFF

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MODULE 11

Campaigns

P1

P1 — High

FEATURES

Campaign CRUD, leads, batches, ops (start/stop), analytics, callbacks, transfers, live queue, batch imports.

TEST SCENARIOS

- [ ] Create a campaign, import leads (batch), and start it.
- [ ] Start/stop/pause controls behave correctly; live queue reflects state.
- [ ] Per-lead call outcomes and campaign analytics aggregate correctly.
- [ ] Scheduled callbacks fire at the right time.
- [ ] Campaign-generated bookings/transfers are tracked.
- [ ] Concurrent-call limits are respected during a campaign.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 12

Knowledge Base

P1

P1 — High

FEATURES

Document upload, FTS search (PostgreSQL, no embeddings), entries, agent attachment, in-call retrieval.

TEST SCENARIOS

- ☐ Upload a document; it is parsed into searchable entries.
- ☐ FTS search returns relevant entries.
- ☐ Agent attached to a KB retrieves correct content during a call.
- ☐ Removing a source cleans up its entries.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 13

Webhooks & Integrations

P2

P2 — Medium

FEATURES

Outbound webhook endpoints, CRM connections, real-time event streaming.

TEST SCENARIOS

- ☐ Configure an outbound webhook endpoint; events deliver with correct payload.
- ☐ Failed deliveries are handled/visible.
- ☐ CRM connection callback completes.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 14

Settings & Account

P1

P1 — High

FEATURES

Account profile, password change, API key regeneration, request timeouts, theme (light/dark/system), Go-Live checklist, webhook guides.

TEST SCENARIOS

- [ ] Update profile fields persist.
- [ ] Change password works and old password no longer logs in.
- [ ] Regenerating the API key invalidates the old one.
- [ ] Theme switching applies and persists.
- [ ] Go-Live checklist reflects real account state.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 15

Compliance / DPDP

P0

P0 — Launch blocker · for legal

FEATURES

DPDP consent disclosure injection, post-call PII redaction (Aadhaar/PAN/phone), retention, admin-triggered data erasure (right to be forgotten).

TEST SCENARIOS

- [ ] DPDP disclosure phrasing is injected into calls where required and never blocks dialing.
- [ ] Post-call PII scan flags/redacts Aadhaar/PAN/phone in transcripts/analytics.
- [ ] Admin data-erasure permanently anonymizes call logs/transcripts/recordings for a given phone number.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 16

Admin Panel

P1

P1 — High · internal

FEATURES

Overview metrics + spike alerts, user management (suspend/reactivate/soft-delete/edit, concurrent-call limits), impersonation (“Login As User”), finance (revenue, transactions, credit/debit/refund), usage monitoring (STT/TTS/LLM, provider breakdown, alerts + alert emails), managed keys, managed numbers, platform health (active sessions, failure categories), prompt-template management, voice whitelist, DPDP erasure, scheduling/GCal oversight.

TEST SCENARIOS

- ☐ Admin login via x-admin-token works; wrong token rejected.
- ☐ User list metrics (balance, spent, calls, minutes, agents) are accurate.
- ☐ Suspend/reactivate/soft-delete a user; suspended user is blocked appropriately.
- ☐ Impersonation issues a single-use, short-lived exchange code and opens the user’s dashboard (requires migration 0012).
- ☐ Credit/debit/refund adjusts the correct wallet by the correct amount.
- ☐ Usage spike alerts fire and alert emails send.
- ☐ Managed keys add/update/mask correctly; managed-key fallback works at call time.
- ☐ Managed-number pool fetch/assign works.
- ☐ Platform health shows active sessions and categorized failure reasons.
- ☐ Prompt-template upload/activate/preview works; voice whitelist add/preview works.
- ☐ Token-optimization kill switch toggles globally.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes

MODULE 17

Cross-Cutting / Non-Functional

P1

P1 — High

FEATURES

Platform-wide quality gates that span every module.

TEST SCENARIOS

- ☐ API error surfacing: failures show user-visible errors (no silent swallowing).
- ☐ Health endpoints (/healthz, /healthz/deep) return healthy.
- ☐ Concurrency: multiple simultaneous calls don’t cross-contaminate sessions.
- ☐ Server restart does not corrupt state; in-flight calls handled gracefully.
- ☐ Mobile/responsive rendering of key dashboard pages.
- ☐ Browser console is free of blocking errors on main pages.
- ☐ Rate limits on public endpoints (booking, webhooks, OTP) reject abuse cheaply.

SIGN-OFF

Tester	Date	Result (Pass/Fail)	Defect ref / Notes